AGRIVIVE / ZABO-CELL

NAGALAND / ZABO-INSPIRED / TWO-ZONE SILT SEPARATOR

SERIES 18 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



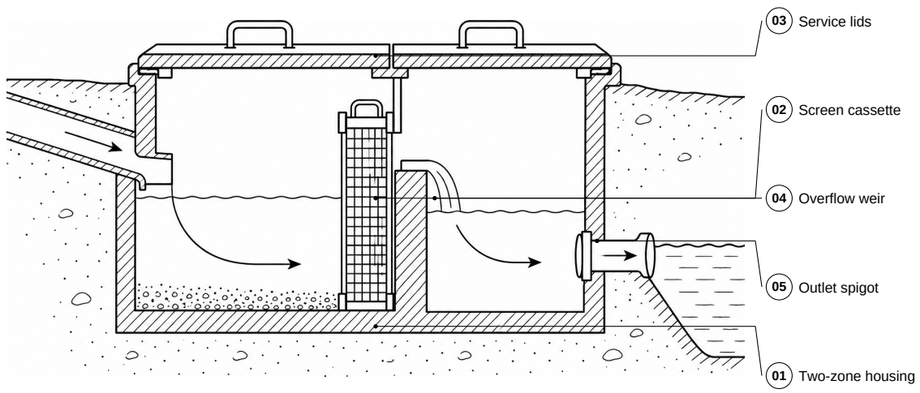
STATE CONTEXT / INDIGENOUS INSPIRATION

Zabo systems in Nagaland integrate catchment management, storage and agricultural water use. ZABO-CELL proposes a removable screen and silt pocket ahead of a small pond; settling performance remains untested.

Practice source: ICAR: Zabo integrated farming system, Nagaland | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 180 L x 110 W x 140 H

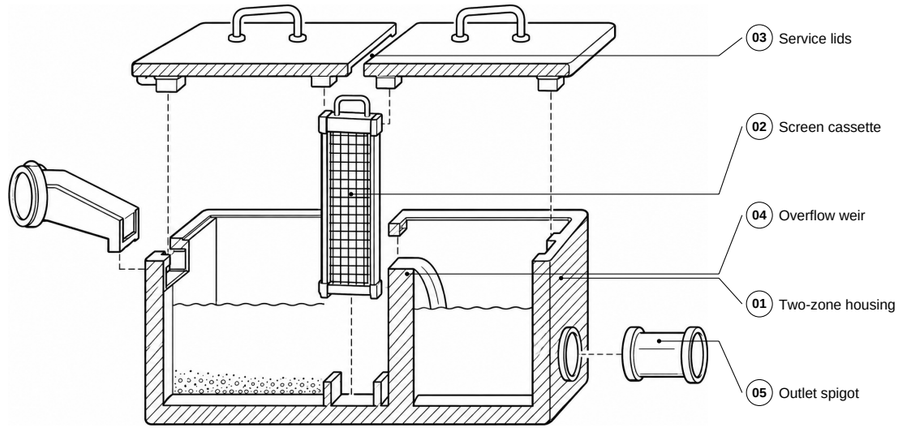


DESIGN / OPERATION NOTES

Seat the housing on a gravity route before pond storage.
Fit the screen between the silt pocket and overflow bay.
Remove the lids for screen cleaning and sediment removal.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
180 L x 110 W x 140 H

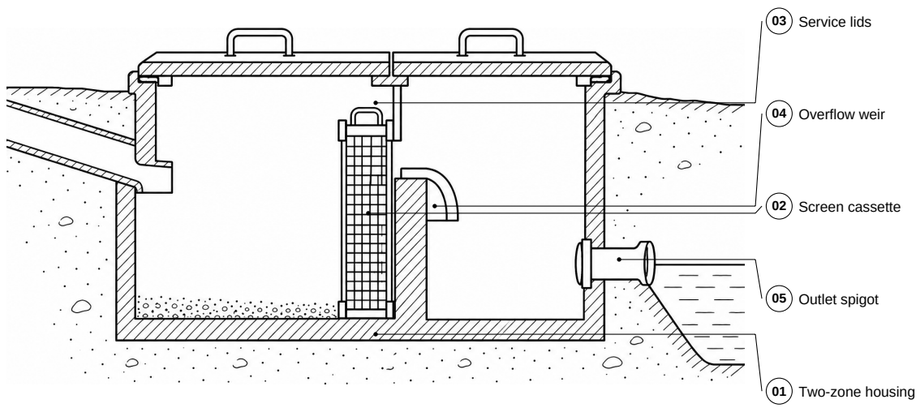


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

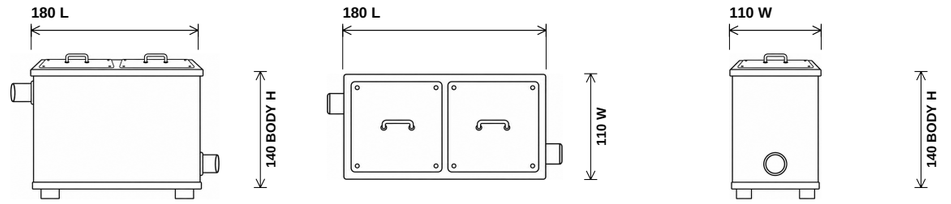
SECTION BASIS / mm
100 L x 90 W x 50 H silt compartment; outlet30 ID



DESIGN / OPERATION NOTES

Fit the screen between the silt pocket and overflow bay.
Remove the lids for screen cleaning and sediment removal.

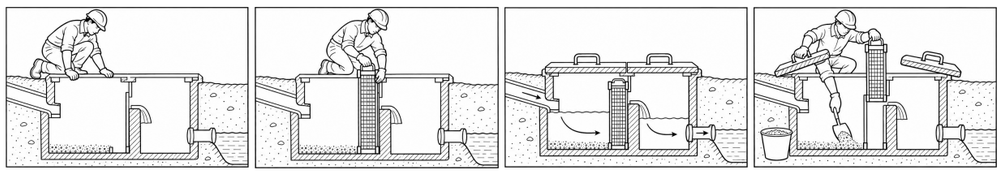
03 ORTHOGRAPHIC STUDIES / mm



LONG ELEVATION **TOP / BODY** **END / BODY**

External 180 L x 110 W x 140 H. Gross pocket 100 L x 90 W x 50 H. Outlet DIA 30 ID.

05 INSTALLATION / OPERATING SEQUENCE



1 SEAT HOUSING **2 FIT SCREEN** **3 CONNECT / COVER** **4 REMOVE SILT**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

External L x W x H	**180 x 110 x 140**
Silt pocket L x W x H	**100 x 90 x 50**
Outlet ID	**30**

02 GEOMETRIC CHECK

GROSS SEDIMENT POCKET

450.0 mL

100 x 90 x 50 / 1000
Working capacity lower; settleability requires trials

03 PARTS / QTY

01 Two-zone housing	**1**
02 Screen cassette	**1**
03 Service lids	**2**
04 Overflow weir	**Integral**
05 Outlet spigot	**Integral**

Body/lids/weir: candidate polymer. Screen: compatible coarse mesh, aperture TBD.

04 INTERFACES / CHECKS

Check short-circuit paths around screen and weir.

Test sediment capture and service interval.

Measure working volume after freeboard and internals.

05 DATUMS / DRAWING BASIS

A: lid seating plane. B: chamber floor. Keep screen withdrawal path clear.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-18 / P01	ZABO-CELL	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	18 / 28